Introduction to Probability and Statistics Fourteenth Edition

Chapter 15
Nonparametric Statistics

Introduction

What are Nonparametric Statistics?

- In all of the preceding chapters we have focused on testing and estimating parameters associated with distributions.
- In this chapter we will focus on questions such as:
  - Do two distributions have the same center?
  - Do two distributions have the same shape?

Why Use Nonparametric Statistics?

- Parametric tests are based upon assumptions that may include the following:
  - The data have the **same variance**, regardless of the treatments or conditions in the experiment.
  - The data are **normally distributed** for each of the treatments or conditions in the experiment.
- What happens when we are not sure that these assumptions have been satisfied?

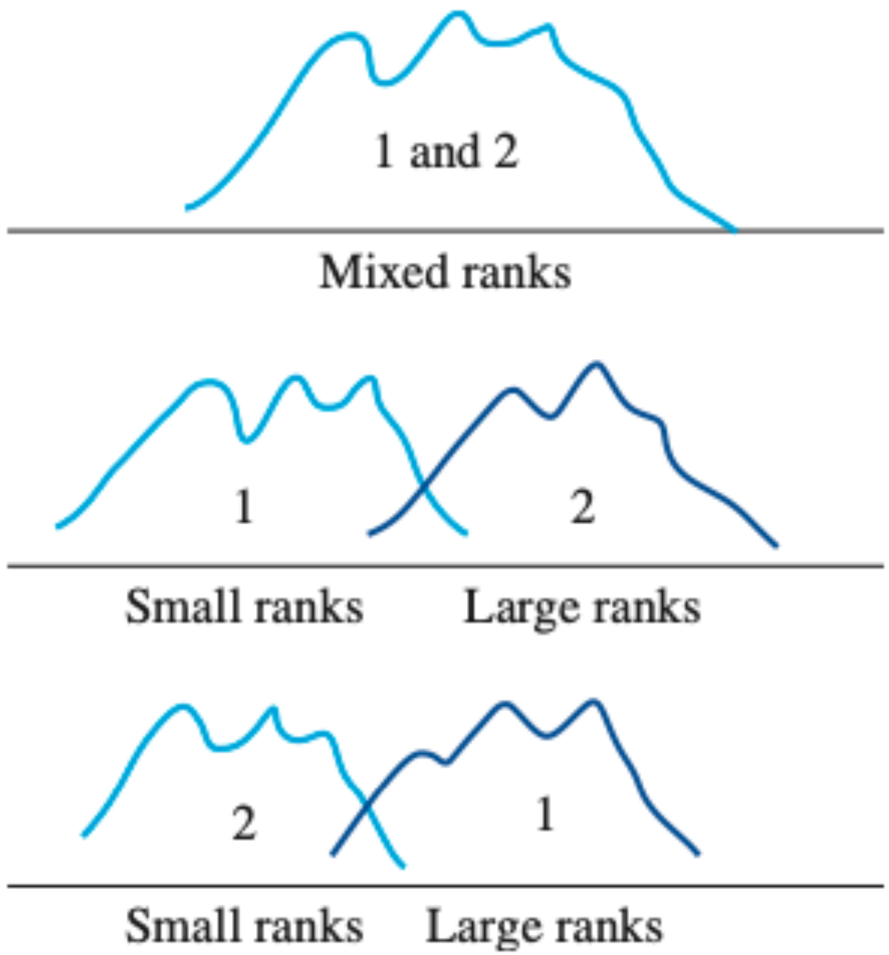
How Do Nonparametric Tests Compare with the Usual *z*, *t*, and *F* Tests?

- Studies have shown that when the usual assumptions are satisfied, nonparametric tests are about 95% efficient when compared to their parametric equivalents.
- When normality and common variance are not satisfied, the nonparametric procedures can be much more efficient than their parametric equivalents.

Extensions to previous materials

Previous	Now
Independent *t* test	15.1 Wilcoxon Rank Sum Test
Paired *t* test	15.2 Sign test
	15.4 Wilcoxon Signed-Rank Test
Completely Randomized Design	15.5 Kruskal-Wallis *H* test
Randomized Block Design	15.6 Freidman F_r test
Person's corr. coeff.	15.7 Spearman's rank corr. coeff.

15.1 THE WILCOXON RANK SUM TEST: INDEPENDENT RANDOM SAMPLES



1 and 2
Mixed ranks
1
2
Small ranks
Large ranks
2
1
Small ranks
Large ranks

The Wilcoxon Rank Sum Test

- Suppose we wish to test the hypothesis that two distributions have the same center.
- We select two independent random samples from each population. Designate each of the observations from population 1 as an "**A**" and each of the observations from population 2 as a "**B**".
- If H_0 is true, and the two samples have been drawn from the same population, when we rank the values in both samples from small to large, **the A's and B's should be randomly mixed in the rankings**.

What happens when H_0 is true?

- Suppose we had 5 measurements from population 1 and 6 measurements from population 2.

- If they were drawn from the same population, the rankings might be like this.

ABABBABABBA

- In this case if we summed the ranks of the A measurements and the ranks of the B measurements, the sums would be similar.

WHAT HAPPENS IF H_0 IS NOT TRUE?

- If the observations come from two different populations, perhaps with population 1 lying to the left of population 2, the ranking of the observations might take the following ordering.

AAABABABBBB

In this case the sum of the ranks of the B observations would be larger than that for the A observations.

Comparing these two scenarios

- What happens when H_0 is true?

ABABBABABBA

- What happens if H_0 is not true?

AAABABABBBB

How to Implement Wilcoxon's Rank Test

- Rank the combined sample from smallest to largest.
- Let T_1 represent the sum of the ranks of the first sample (**A**'s).
- Then, T_1^* defined below, is the sum of the ranks that the A's would have had if the observations were ranked from *large to small.*

$$\boxed{T_1^* = n_1(n_1 + n_2 + 1) - T_1}$$

The Wilcoxon Rank Sum Test

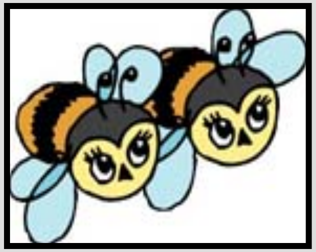
H_0: the two population distributions are the same

H_a: the two populations are in some way different

- The **test statistic** is the smaller of T_1 and T_1^*.
- Reject H_0 if the test statistic is less than the **critical value** found in Table 7(a).
- Table 7(a) is indexed by letting population 1 be the one associated with the smaller sample size n_1, and population 2 as the one associated with n_2, the larger sample size.

Example 15.1. i

THE BEE PROBLEM



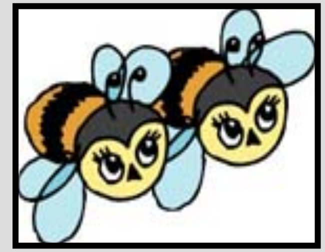
The wing stroke frequencies of two species of bees were recorded for a sample of $n_1 = 4$ species 1 bees and $n_2 = 6$ species 2 bees.[1] The frequencies are listed in Table 15.3. Can you conclude that the distributions of wing strokes differ for these two species? Test using $\alpha = .05$.

Table 15.3 Wing Stroke Frequencies for Two Species of Bees

Species 1	Species 2
235	180
225	169
190	180
188	185
	178
	182

H0: distributions of wing strokes are equal ($M_1 = M_2$)
H1: distributions of wing strokes differ for these two species ($M_1 \neq M_2$)

Example 15.1. ii



1. We rank the 10 observations from smallest to largest, Table 15.4.
2. The sample with the smaller sample size is called sample 1.

Solution You first need to rank the observations from small to large, as shown in Table 15.4.

Table 15.4 Wing Stroke Frequencies Ranked from Small to Large

Data	Species	Rank
169	2	1
178	2	2
180	2	3
180	2	4
182	2	5
185	2	6
188	1	7
190	1	8
225	1	9
235	1	10

$n_1 = 4$, $n_2 = 6$. Therefore,

$$T_1 = 7 + 8 + 9 + 10 = 34,$$

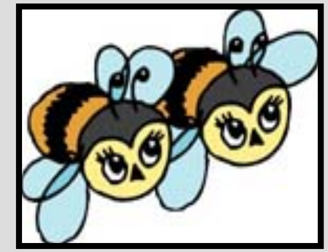
and

$$T^* = n_1(n_1 + n_2 + 1) - T_1 = 4(4 + 6 + 1) - 34 = 10.$$

Finally, the statistic is

$$T = \min(T_1, T^*) = \min(34,10) = 10$$

Example 15.1. iii



$n_1 = 4$, $n_2 = 6$. Therefore,

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$$T^* = n_1(n_1 + n_2 + 1) - T_1 = 4(4 + 6 + 1) - 34 = 10.$$

Finally, the statistic is

$$T = \min(T_1, T^*) = \min(34, 10) = 10$$

The Bee Problem

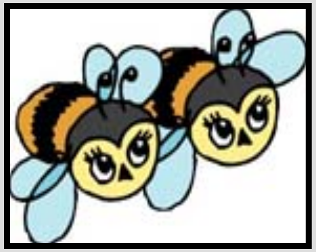


TABLE 7(b)
2.5% Left-Tailed Critical Values

	n_1							
n_2	2	3	4	5	6	7	8	9
4	—	—	10					
5	—	6	11	17				
6	—	7	12	18	26			
7	—	7	13	20	27	36		
8	3	8	14	21	29	38	49	
9	3	8	14	22	31	40	51	62

Calculate $T_1 = 7 + 8 + 9 + 10 = 34$

$$T_1^* = n_1(n_1 + n_2 + 1) - T_1$$
$$= 4(4 + 6 + 1) - 34 = 10$$

1. The test statistic is T = 10.

2. The critical value of T from Table 7(b) for a two-tailed test with $\alpha/2 = .025$ is T = 12; H_0 is rejected if $T \leq 12$.

Reject H_0. Sufficient evidence to indicate a difference in the distributions of wing stroke frequencies.

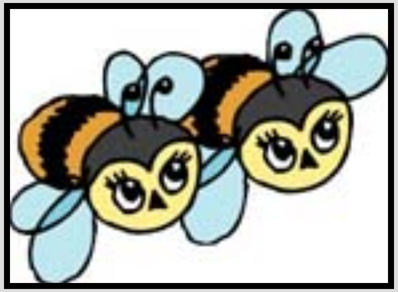
This is originally a two-tailed test. However, by the definition of $T = \min(T_1, T^*)$, we only need to focus on the left tail.

Use Table 7(b) in Appendix I with $n_1 = 4$ and $n_2 = 6$, and the critical value of T such that $P(T \leq a) = \alpha/2 = 0.025$ is 12.

Therefore, you should reject H_0 if the observed value of T is 12 or less. Because $T = 10$, we reject H_0.

MINITAB OUTPUT

Recall $T_1 = 34; T_1^* = 10.$



Mann-Whitney Test and CI: Species1, Species2

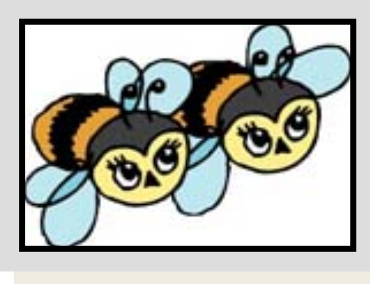
```
              N   Median
Species 1    4   207.50
Species 2    6   180.00

Point estimate for ETA1-ETA2 is 30.50
95.7 Percent CI for ETA1-ETA2 is (5.99,56.01)
W = 34.0
Test of ETA1 = ETA2 vs ETA1 not = ETA2 is significant at
0.0142
The test is significant at 0.0139 (adjusted for ties)
```

Minitab calls the procedure the Mann-Whitney U Test, equivalent to the Wilcoxon Rank Sum Test.

The test statistic is W = T_1 = 34 and has *p*-value = .0142. Reject H_0 for α = .05.

Large Sample Approximation: Wilcoxon Rank Sum Test



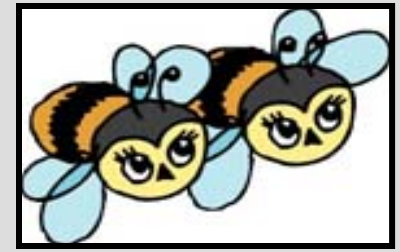
When n_1 and n_2 are large (greater than 10 is large enough), a normal approximation can be used to approximate the critical values in Table 7.

1. Calculate T_1 and T_1^*. Let $T = \min(T_1, T_1^*)$.

2. The statistic $z = \dfrac{T - \mu_T}{\sigma_T}$ has an approximate z distribution with

$$\mu_T = \frac{n_1(n_1 + n_2 + 1)}{2} \text{ and } \sigma_T^2 = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

Some Notes



- When should you use the Wilcoxon Rank Sum test instead of the two-sample *t* test for independent samples?
  - ✓ when a normality plot shows a **violation of the normality assumption**
  - ✓ when the responses can only be **ranked** and not quantified
  - ✓ when the F test or the Rule of Thumb shows a **problem with equality of variances (ANOVA: CRD with 2 treatments)**

15.2 the Sign test for a paired experiment

THE SIGN TEST

- The **sign test** is a fairly simple procedure that can be used to compare two populations when the samples consist of **paired observations**.
- It can be used
  - ✓ when the assumptions required for the **paired-difference test** of Chapter 10 are not valid or
  - ✓ when the responses can only be ranked as "**one better than the other**", but cannot be quantified.

Bernoulli distribution

- (A is better than B) $\sim$ Bernoulli(p=0.5)
- In an experiment with n paired observations, we will use Binomial$(n, p = 0.5)$

The Sign Test

✓For each pair, measure whether the first response—say, A—exceeds the second response—say, B.

✓The test statistic is ***x***, the number of times that A exceeds B in the *n* pairs of observations.

✓Only pairs without ties are included in the test.

✓Critical values for the rejection region or exact *p*-values can be found using the cumulative binomial tables in Appendix I.

The Sign Test

H_0: the two populations are identical versus

H_a: one or two-tailed alternative

is equivalent to

H_0: p = P(A exceeds B) = .5   versus

H_a: p ($\neq$, <, or >) .5

Test statistic: ***x* = number of plus signs**

Rejection region or *p*-values from Table 1.

Example

Two gourmet chefs each tasted and rated eight different meals from 1 to 10. Does it appear that one of the chefs tends to give higher ratings than the other? Use $\alpha = .01$.

Meal	1	2	3	4	5	6	7	8
Chef A	6	4	7	8	2	4	9	7
Chef B	8	5	4	7	3	7	9	8
Sign	-	-	+	+	-	-	0	-

H_0: the rating distributions are the same ($p = .5$)

H_a: the ratings are different ($p \neq .5$)

The Gourmet Chefs

Meal	1	2	3	4	5	6	7	8
Chef A	6	4	7	8	2	4	9	7
Chef B	8	5	4	7	3	7	9	8
Sign	-	-	+	+	-	-	0	-

p-value =.454 is too large to reject H_0. There is insufficient evidence to indicate that one chef tends to rate one meal higher than the other.

H_0: $p = .5$

H_a: $p \neq .5$ with $n = 7$ (omit the tied pair)

Test Statistic: x = number of plus signs = 2

Use Table 1 with n = 7 and p = .5.

p-value = P(observe x = 2 or something equally as unlikely)

= $P(x \leq 2) + P(x \geq 5) = 2(.227) = .454$

k	0	1	2	3	4	5	6	7
$P(x \leq k)$	.008	.062	.227	.500	.773	.938	.992	1.000

REWRITE THE ANSWER USING OUR PROCEDURES

1. $H_0: \ p = 0.5$ vs $H_a: p \neq 0.5$

2. $\alpha = 0.01$

3. Test statistic

$$x = \#(+) \sim Binomial(7,0.5)$$

4. Realized statistic: $x^* = 2$

5. $p-$value $= 2 \times P(X \leq 2) = 2 \times 0.227 = 0.454$

6. Do not reject H0.

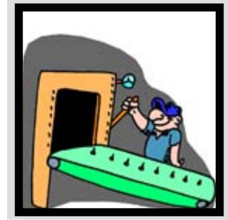
Large Sample Approximation: The Sign Test

When $n \geq 25$, a normal approximation can be used to approximate the critical values in Table 1.

1. Calculate x = number of plus signs.

2. The statistic $z = \dfrac{x - .5n}{.5\sqrt{n}}$ has an approximate z distribution. Rejection regions and p-values can be found using Table 3.

Example



You record the number of accidents per day at a large manufacturing plant for both the day and evening shifts for $n = 100$ days. You find that the number of accidents per day for the evening shift x_E exceeded the corresponding number of accidents in the day shift x_D on 63 of the 100 days. Do these results provide sufficient evidence to indicate that more accidents tend to occur on one shift than on the other?

H_0: the distributions (# of accidents) are the same ($p = .5$)

H_a: the distributions are different ($p \neq .5$)

$$\text{Test statistic: } z = \frac{x - .5n}{.5\sqrt{n}} = \frac{63 - .5(100)}{.5\sqrt{100}} = 2.60$$

For a two tailed test, we reject H_0 if $|z| > 1.96$ (5% level).
H_0 is rejected. There is evidence of a difference between the day and night shifts.

Example

You record the number of accidents per day at a large manufacturing plant for both the day and evening shifts for n = 100 days. You find that the number of accidents per day for the evening shift x_E exceeded the corresponding number of accidents in the day shift x_D on 63 of the 100 days. Do these results provide sufficient evidence to indicate that more accidents tend to occur on one shift than on the other?

Solution

1. Set up

   H_0 : the distributions (numberof accidents) are the same $\quad (p = .5)$

   H_a : the distributions are different $\quad (p \neq .5)$

2. Set $\alpha = 5\%$.
3. The test statistic
$$Z_{STAT} = \frac{x - .5n}{.5\sqrt{n}} \sim Z.$$
4. The realized test statistic
$$Z^* = \frac{63 - .5 \times 100}{.5\sqrt{100}} = 2.60.$$
5. For a two tailed test, the rejection region is $\{z : |z| > 1.96\}$ (5% level).
6. H_0 is rejected. There is evidence of a difference between the day and night shifts.

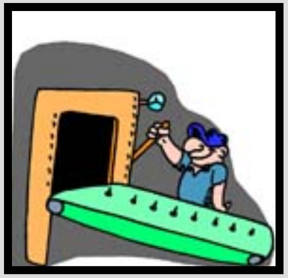
15.3 A COMPARISON OF STATISTICAL TESTS

WHICH TEST SHOULD YOU USE?

- We compare statistical tests using
  Definition: Power = $1 - \beta$

  $$= P(\text{reject } H_0 \mid H_0 \text{ is false})$$

- The **power** of the test is the probability of rejecting the null hypothesis when it is false and some specified alternative is true.
- The **power** is the probability that the test will do what it was designed to do—that is, detect a departure from the null hypothesis when a departure exists.



- If all parametric assumptions have been met, the parametric test will be the most powerful.
- If not, a nonparametric test may be more powerful.
- If you can reject H_0 with a less powerful nonparametric test, you will not have to worry about parametric assumptions.
- If not, you might try
  - more powerful nonparametric test or
  - increasing the sample size to gain more power

- If you try a test and can reject H_0, then you are happy and stop.
- If you try a test but can not reject H_0, then you should try some other tests.
- See "20230515 stat 09.2 power.pdf" for more details.

15.4 THE WILCOXON SIGNED-RANK TEST FOR A PAIRED EXPERIMENT

THE WILCOXON SIGNED-RANK TEST

- The **Wilcoxon Signed-Rank Test** is a more powerful nonparametric procedure that can be used to compare two populations when the samples consist of **paired observations.**

- It uses the **ranks** of the differences, $d = x_1 - x_2$ that we used in the paired-difference test.

The Wilcoxon Signed-Rank Test

- ✓For each pair, calculate the difference $d = x_1 - x_2$. Eliminate zero differences.
- ✓Rank the absolute values of the differences from 1 to n. Tied observations are assigned average of the ranks they would have gotten if not tied.
  - **T^+ = rank sum for positive differences**
  - **T^- = rank sum for negative differences**
- ✓If the two populations are the same, T^+ and T^- should be nearly equal. If either T^+ or T^- is unusually large, this provides evidence against the null hypothesis.

The Wilcoxon Signed-Rank Test

H_0: the two populations are identical

versus

H_a: one or two-tailed alternative

Test statistic: **T = smaller of T^+ and T^-**

Critical values for a one or two-tailed rejection region can be found using **Table 8** in Appendix I.

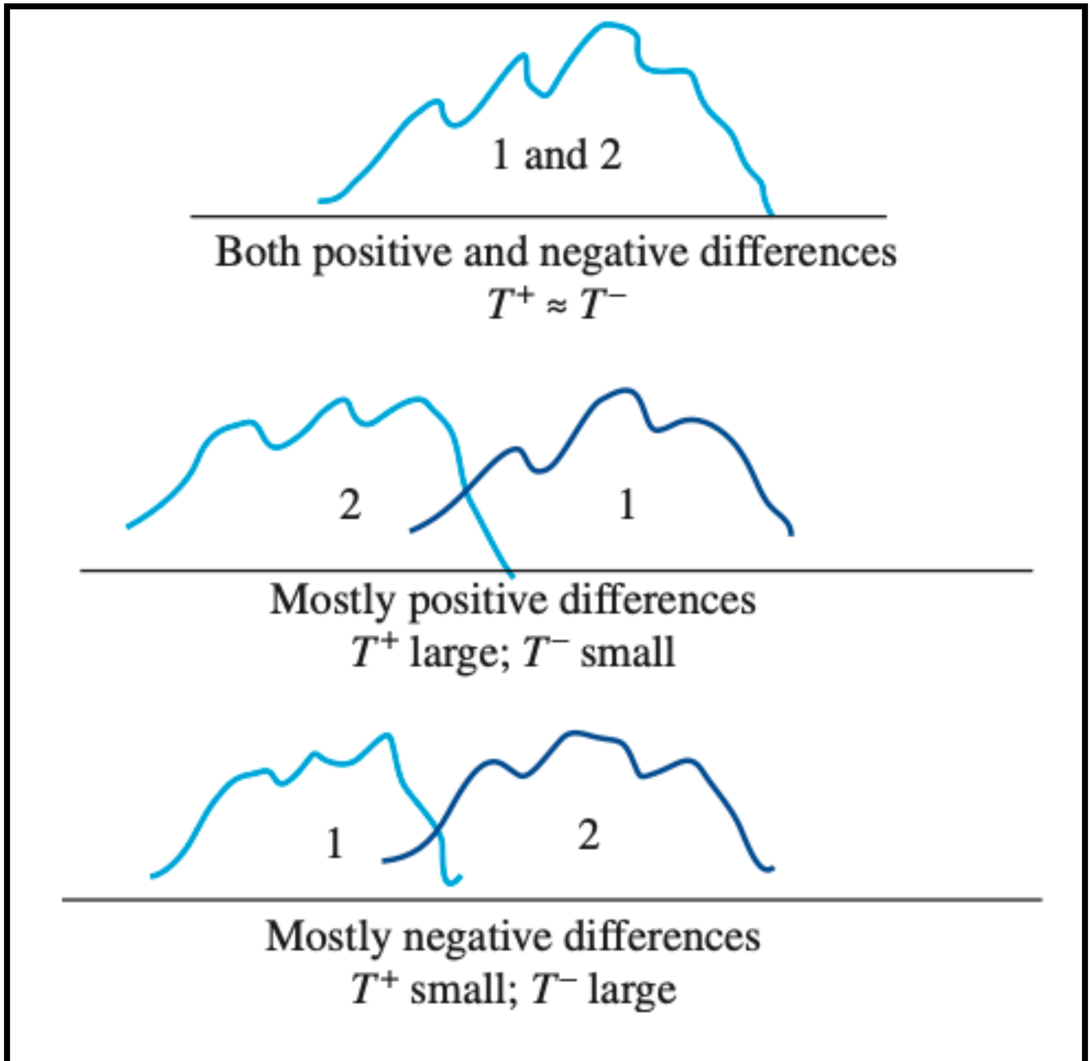

1 and 2
Both positive and negative differences
$T^+ \approx T^-$
2
1
Mostly positive differences
T^+ large; T^- small
1
2
Mostly negative differences
T^+ small; T^- large

Table 15.7 An Abbreviated Version of Table 8 in Appendix I; Critical Values of T

One-Sided	Two-Sided	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$	$n = 10$	$n = 11$
$\alpha = .050$	$\alpha = .10$	1	2	4	6	8	11	14
$\alpha = .025$	$\alpha = .05$		1	2	4	6	8	11
$\alpha = .010$	$\alpha = .02$			0	2	3	5	7
$\alpha = .005$	$\alpha = .01$				0	2	3	5

One-Sided	Two-Sided	$n = 12$	$n = 13$	$n = 14$	$n = 15$	$n = 16$	$n = 17$
$\alpha = .050$	$\alpha = .10$	17	21	26	30	36	41
$\alpha = .025$	$\alpha = .05$	14	17	21	25	30	35
$\alpha = .010$	$\alpha = .02$	10	13	16	20	24	28
$\alpha = .005$	$\alpha = .01$	7	10	13	16	19	23

Figure 15.2
Rejection region for the Wilcoxon signed-rank test for a paired experiment (reject H_0 if $T \le T_0$)

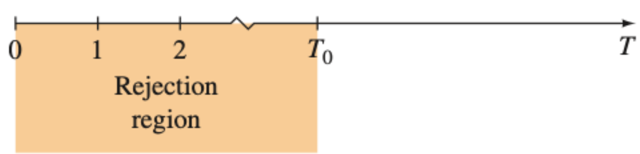


Table 8 Critical Values of T for the Wilcoxon Signed-Rank Test, $n = 5(1)50$

One-Sided	Two-Sided	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$	$n = 10$
$\alpha = .050$	$\alpha = .10$	1	2	4	6	8	11
$\alpha = .025$	$\alpha = .05$		1	2	4	6	8
$\alpha = .010$	$\alpha = .02$			0	2	3	5
$\alpha = .005$	$\alpha = .01$				0	2	3

One-Sided	Two-Sided	$n = 11$	$n = 12$	$n = 13$	$n = 14$	$n = 15$	$n = 16$
$\alpha = .050$	$\alpha = .10$	14	17	21	26	30	36
$\alpha = .025$	$\alpha = .05$	11	14	17	21	25	30
$\alpha = .010$	$\alpha = .02$	7	10	13	16	20	24
$\alpha = .005$	$\alpha = .01$	5	7	10	13	16	19

One-Sided	Two-Sided	$n = 17$	$n = 18$	$n = 19$	$n = 20$	$n = 21$	$n = 22$
$\alpha = .050$	$\alpha = .10$	41	47	54	60	68	75
$\alpha = .025$	$\alpha = .05$	35	40	46	52	59	66
$\alpha = .010$	$\alpha = .02$	28	33	38	43	49	56
$\alpha = .005$	$\alpha = .01$	23	28	32	37	43	49

One-Sided	Two-Sided	$n = 23$	$n = 24$	$n = 25$	$n = 26$	$n = 27$	$n = 28$
$\alpha = .050$	$\alpha = .10$	83	92	101	110	120	130
$\alpha = .025$	$\alpha = .05$	73	81	90	98	107	117
$\alpha = .010$	$\alpha = .02$	62	69	77	85	93	102
$\alpha = .005$	$\alpha = .01$	55	68	68	76	84	92

One-Sided	Two-Sided	$n = 29$	$n = 30$	$n = 31$	$n = 32$	$n = 33$	$n = 34$
$\alpha = .050$	$\alpha = .10$	141	152	163	175	188	201
$\alpha = .025$	$\alpha = .05$	127	137	148	159	171	183
$\alpha = .010$	$\alpha = .02$	111	120	130	141	151	162
$\alpha = .005$	$\alpha = .01$	100	109	118	128	138	149

Example

To compare the densities of cakes using mixes A and B, six pairs of pans (A and B) were baked side-by-side in six different oven locations. Is there evidence of a difference in density for the two cake mixes?

Location	1	2	3	4	5	6
Cake Mix A	.135	.102	.098	.141	.131	.144
Cake Mix B	.129	.120	.112	.152	.135	.163
$d = x_A - x_B$	.006	-.018	-.014	-.011	-.004	-.019

H_0: the density distributions are the same
H_a: the density distributions are different

Cake Densities

Location	1	2	3	4	5	6
Cake Mix A	.135	.102	.098	.141	.131	.144
Cake Mix B	.129	.120	.112	.152	.135	.163
$d = x_A - x_B$	.006	-.018	-.014	-.011	-.004	-.019
Rank	2	5	4	3	1	6

Rank the 6 differences, without regard to sign.

Do not reject H_0. There is insufficient evidence to indicate that there is a difference in densities for the two cake mixes.

Calculate: $T^+ = 2$ and $T^- = 5+4+3+1+6 = 19$.
The **test statistic** is $T = 2$.
Rejection region: Use Table 8. For a two-tailed test with $\alpha = .05$, reject H_0 if $T \leq 1$.

TABLE 8
Critical Values of T in the Wilcoxon Signed-Rank Test, $n = 5(1)50$

One-Sided	Two-Sided	$n = 5$	$n = 6$	$n = 7$
$\alpha = .050$	$\alpha = .10$	1	2	4
$\alpha = .025$	$\alpha = .05$		1	2
$\alpha = .010$	$\alpha = .02$			0
$\alpha = .005$	$\alpha = .01$			

LARGE SAMPLE APPROXIMATION: THE SIGNED-RANK TEST

When $n \geq 25$, a normal approximation can be used to approximate the critical values in Table 8.

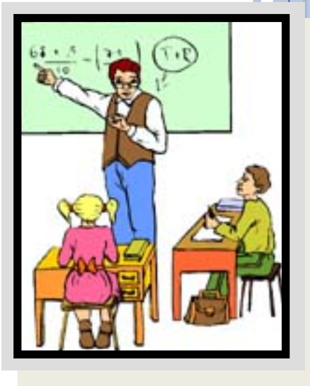
1. Calculate T^+ and T^-. Let $T = \min(T^+, T^-)$.

2. The statistic $z = \dfrac{T - \mu_T}{\sigma_T}$ has an approximate z distribution with

$$\mu_T = \frac{n(n+1)}{4} \text{ and } \sigma_T^2 = \frac{n(n+1)(2n+1)}{24}$$

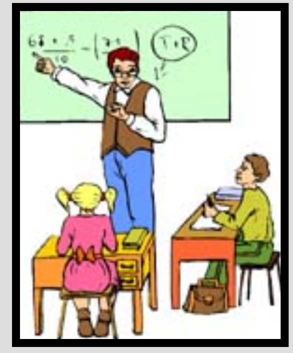
15.5 the Kruskal–Wallis H-test for completely randomized designs

The Kruskal-Wallis *H* Test



- The **Kruskal-Wallis *H* Test** is a nonparametric procedure that can be used to compare more than two populations in a **completely randomized design.**
- All $n = n_1+n_2+\ldots+n_k$ measurements are jointly ranked.
- We use the sums of the ranks of the k samples to compare the distributions.

THE KRUSKAL-WALLIS *H* TEST



✓Rank the total measurements in all *k* samples from 1 to *n*. Tied observations are assigned average of the ranks they would have gotten if not tied.

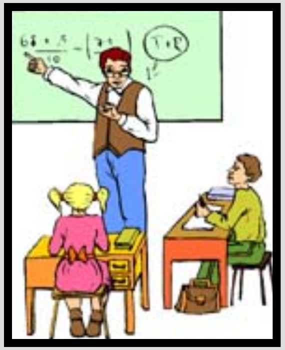
✓Calculate

- **T_i = rank sum for the *i*th sample $i = 1, 2, \dots, k$**

✓And the test statistic

$$H = \frac{12}{n(n+1)} \sum \frac{T_i^2}{n_i} - 3(n+1)$$

THE KRUSKAL-WALLIS *H* TEST



H_0: the *k* distributions are identical versus

H_a: at least one distribution is different

Test statistic: ***Kruskal-Wallis H***

When H_0 is true, the test statistic *H* has an approximate chi-square distribution with *df* = (*k*-1).

Use a right-tailed rejection region or *p*-value based on the Chi-square distribution.

The Fruskal-Wallis H test

1. H_0: The k distributions are identical versus H_a: at least one distribution is different
2. Set α.
3. Test statistic and its distribution

$$H_{STAT} = \boxed{\frac{12}{n(n+1)} \sum \frac{T_i^2}{n_i} - 3(n+1)} \sim \chi^2_{(k-1)}$$

4. Calculate H^*.
5. Right-tailed test. Find rejection region or p-value.
6. Conclude.

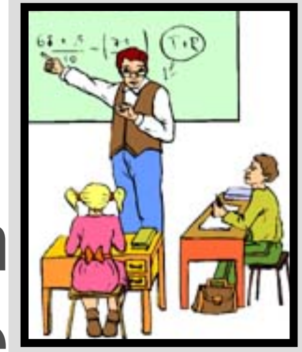
Why is this a right-tailed test?

- If H0 is true
- k distributions are the same
- The realized $T_1, \ldots, T_k$ will be about the same
- The realized H will be close to zero.

- If H0 is false
- At least one distribution is diffrent
- At least realized $T_1, \ldots, T_k$ will be very different from the others
- The realized H will be large.

Recall in ANOVA of ch 11, we also have a right-tailed test.

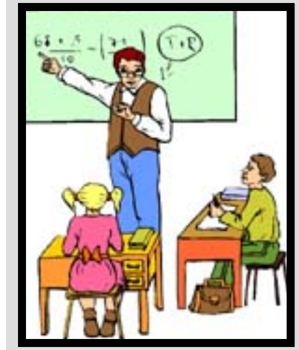
Example



Four groups of students were random assigned to be taught with four differe techniques, and their achievement test scores were recorded. Are the distributions of test scores the same, or do they differ in location?

1	2	3	4
65	75	59	94
87	69	78	89
73	83	67	80
79	81	62	88

Teaching Methods



1	2	3	4
65	75	59	94
87	69	78	89
73	83	67	80
79	81	62	88

	(3)	(7)	(1)	(16)
	(13)	(5)	(8)	(15)
	(6)	(12)	(4)	(10)
	(9)	(11)	(2)	(14)
T_i	31	35	15	55

Rank the 16 measurements from 1 to 16, and calculate the four rank sums.

H_0: the distributions of scores are the same

H_a: the distributions differ in location

$$\text{Test statistic: } H = \frac{12}{n(n+1)} \sum \frac{T_i^2}{n_i} - 3(n+1)$$

$$= \frac{12}{16(17)} \left(\frac{31^2 + 35^2 + 15^2 + 55^2}{4} \right) - 3(17) = 8.96$$

Calculation

Raw data

1	2	3	4
65	75	59	94
87	69	78	89
73	83	67	80
79	81	62	88

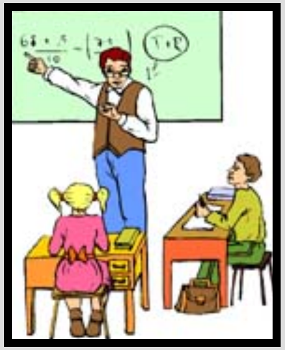
Rank data

1	2	3	4
3	7	1	16
13	5	8	15
6	12	4	10
9	11	2	14

Ti	31	35	15	55
Ti^2	961	1225	225	3025
ni	4	4	4	4
Ti^2/n_i	240.25	306.25	56.25	756.25
n	16			

H	8.9558824

Teaching Methods



H_0: the distributions of scores are the same
H_a: the distributions differ in location

$$\text{Test statistic: } H = \frac{12}{n(n+1)}\sum \frac{T_i^2}{n_i} - 3(n+1)$$

$$= \frac{12}{16(17)}\left(\frac{31^2 + 35^2 + 15^2 + 55^2}{4}\right) - 3(17) = 8.96$$

Rejection region: Use Table 5. For a right-tailed chi-square test with α = .05 and *df* = 4-1 =3, reject H_0 if $H \geq 7.81$.

Reject H_0. There is sufficient evidence to indicate that there is a difference in test scores for the four teaching techniques.

Summary

1. H_0: the distributions of scores using four different techniques are the same versus H_0: at least one distribution is different
2. $\alpha = 0.05$
3. Kruskal-Wallis H test statistic $\sim \chi^2_3$
4. $H^* = 8.96$
5. Rejection region $\{H : H > 7.81\}$.
6. Reject H_0. There is sufficient evidence to indicate that there is a difference in test scores for the four teaching techniques.

15.6 the Friedman F_r-test for randomized block designs

THE FRIEDMAN F_R TEST

- The **Friedman F_r Test** is the nonparametric equivalent of the **randomized block design** with k treatments and b blocks.
- All k measurements within a block are ranked from 1 to b.
- We use the sums of the ranks of the k treatment observations to compare the k treatment distributions.

The Friedman F_r test

1. H_0: The k treatments' distributions are identical versus H_a: at least one distribution is different
2. Set α.
3. Test statistic and its distribution

$$F_{r,STAT} = \boxed{\frac{12}{bk(k+1)} \sum T_i^2 - 3b(k+1)} \sim \chi^2_{(k-1)}$$

4. Calculate F_r^*.
5. Right-tailed test. Find rejection region or p-value.
6. Conclude.

Why is this a right-tailed test?

- If H_0 is true
- k distributions are the same
- The realized $T_1, \ldots, T_k$ will be about the same
- The F_r^* will be close to zero.

- If H_0 is false
- At least two treatment distributions are diffrent
- At least realized $T_1, \ldots, T_k$ will be very different from the others
- The F_r^* will be large.

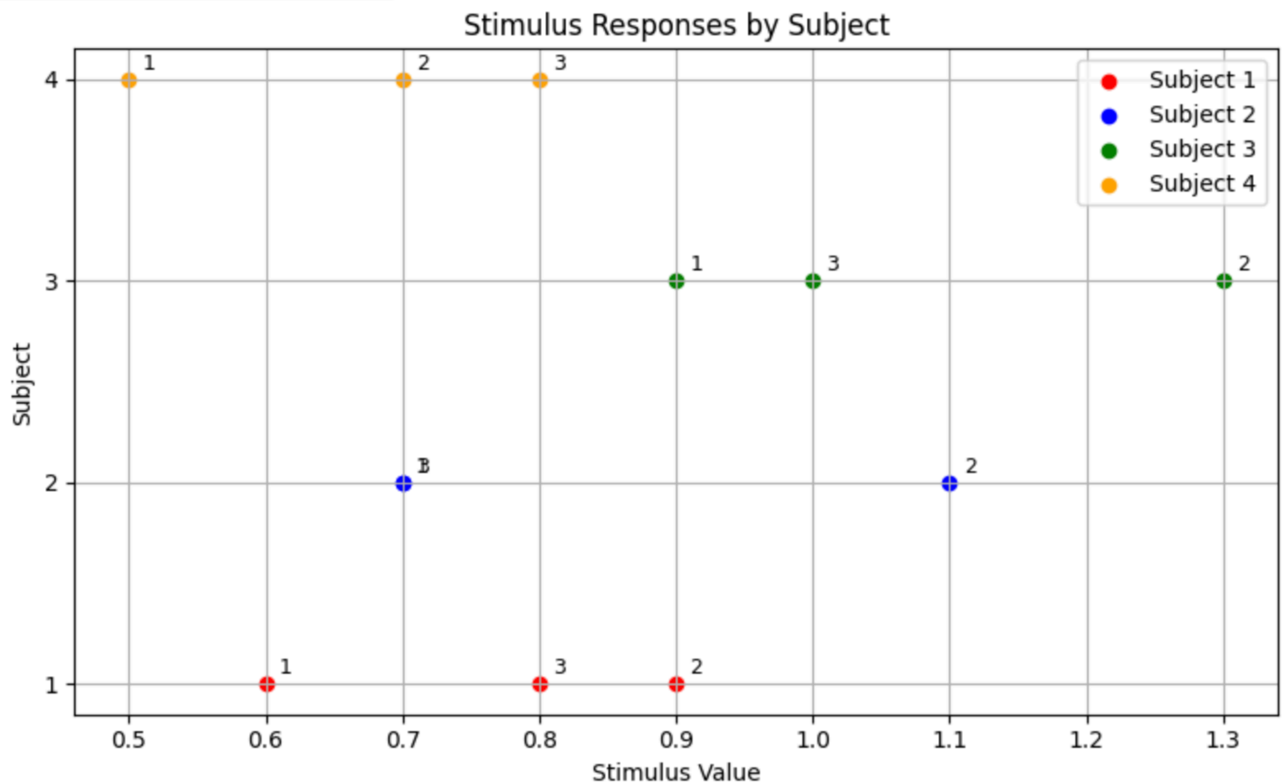
Recall in ANOVA of ch 11, and Kruskal-Wallis H test, we also have a right-tailed test.

Example

A student is subjected to a stimulus and we measure the time until the student reacts by pressing a button. Four students are used in the experiment, each is subjected to three stimuli, and their reaction times are measured. Do the distributions of reaction times differ for the three stimuli?

	Stimuli		
Subject	**1**	**2**	**3**
1	.6	.9	.8
2	.7	1.1	.7
3	.9	1.3	1.0
4	.5	.7	.8

	Stimuli		
Subject	**1**	**2**	**3**
1	.6	.9	.8
2	.7	1.1	.7
3	.9	1.3	1.0
4	.5	.7	.8



https://colab.research.google.com/drive/14uCeFKz-SCct5_0SQm2VtqvvldxYAJ5U?usp=sharing

Reaction Times

	Stimuli		
Subject	1	2	3
1	.6	.9	.8
2	.7	1.1	.7
3	.9	1.3	1.0
4	.5	.7	.8

	(1)	(3)	(2)
	(1.5)	(3)	(1.5)
	(1)	(3)	(2)
	(1)	(2)	(3)
T_i	4.5	11	8.5

Rank the 3 measurements for each subject from 1 to 3, and calculate the three rank sums.

H_0: the distributions of reaction times are the same

H_a: the distributions differ in location

$$\text{Test statistic: } F_r = \frac{12}{bk(k+1)} \sum T_i^2 - 3b(k+1)$$

$$= \frac{12}{12(4)}(4.5^2 + 11^2 + 8.5^2) - 3(4)(4) = 5.375$$

Reaction Times

H_0: the distributions of reaction times are the same
H_a: the distributions differ in location

$$\text{Test statistic: } F_r = \frac{12}{bk(k+1)}\sum T_i^2 - 3b(k+1)$$

$$= \frac{12}{12(4)}(4.5^2 + 11^2 + 8.5^2) - 3(4)(4) = 5.375$$

Rejection region: Use Table 5. For a right-tailed chi-square test with α = .05 and *df* = 3-1 =2, reject H_0 if $H \geq 5.99$.

Do not reject H_0. There is insufficient evidence to indicate that there is a difference in reaction times for the three stimuli.

15.7 Rank correlation Coefficient

EXAMPLE

Five elementary school science teachers have been ranked by a judge according to their teaching ability. They have also taken a national "teacher's exam". Is there agreement between the judge's rank and the exam score?

Teacher	**1**	**2**	**3**	**4**	**5**
Judge's Rank	4	2	3	1	5
Exam score	72	69	82	93	80

If the judge's rank is low (best teacher), we might expect the teacher's score to be high. We look for a negative association between the ranked measurements.

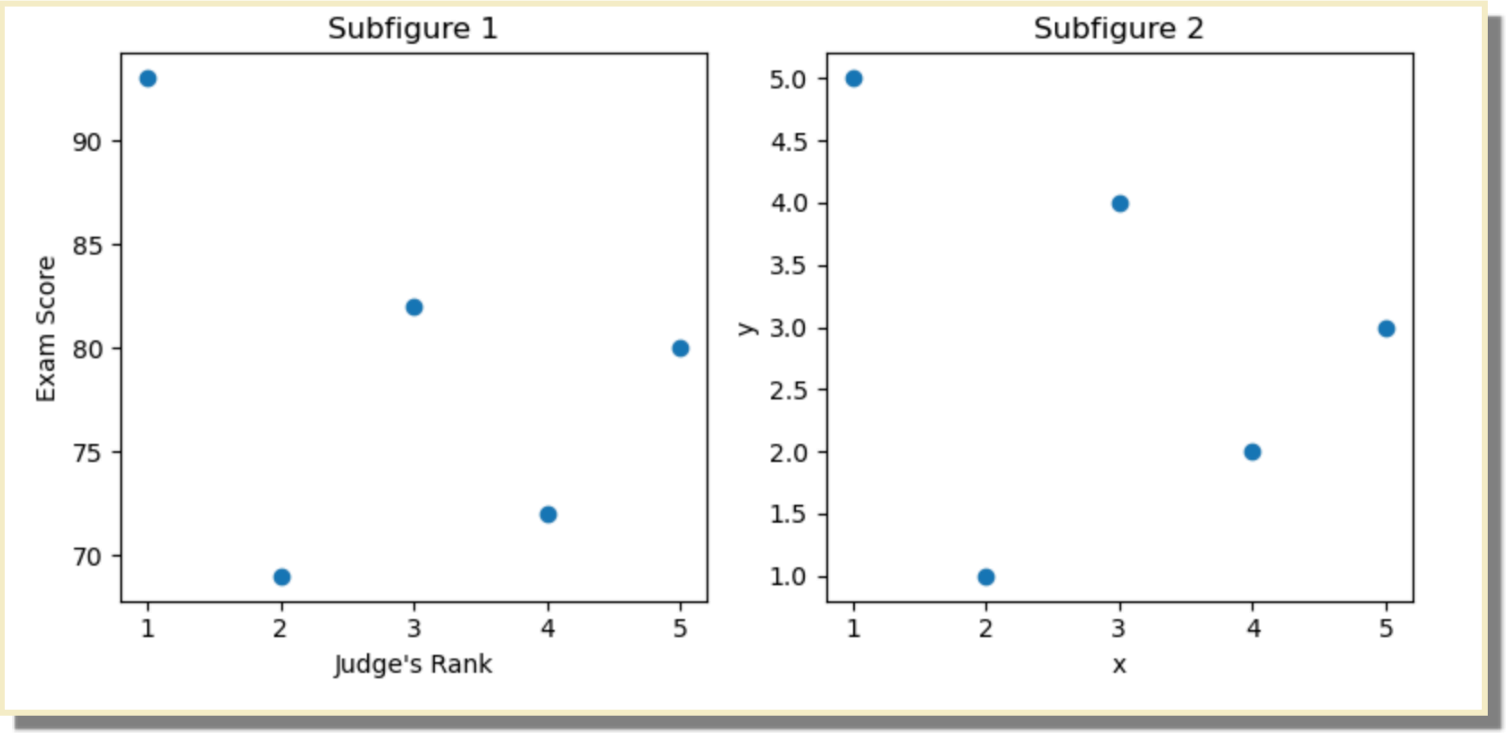
Rank Correlation Coefficien

- The **rank correlation coefficient**, **Spearman** $\boldsymbol{r_s}$ is the nonparametric equivalent of the Pearson correlation coefficient *r* from Chapters 3 and 12.
- The two variables are each ranked from smallest to largest and the **ranks are denoted as *x* and *y*.**
- We are interested in the strength of the relationship (correlation) between the two variables.
- Pearson's "correlation coefficient" measures linear relationship between x and y.
- Spearman's "rank correlation coefficient" measures monotone relationship between x and y. (Monotone: if x increases, y increases. Or, if x increases, y decreases.)

Rank the exam scores (the first variable is already in rank form). There are no ties.

Teacher	1	2	3	4	5
Judge's Rank	4	2	3	1	5
Exam score	72	69	82	93	80

Teacher	1	2	3	4	5
x	4	2	3	1	5
y	2	1	4	5	3



Spearman's rank correlation

Calculate:

$$S_{xy} = \sum(x-\bar{x})(y-\bar{y}) = \sum xy - \frac{\sum x \sum y}{n},$$

$$S_{xx} = \sum(x-\bar{x})^2 = \sum x^2 - \frac{(\sum x)^2}{n},$$

$$S_{yy} = \sum(y-\bar{y})^2 = \sum y^2 - \frac{(\sum y)^2}{n}.$$

Spearman's rank correlation coefficient is

$$r_{s,STAT} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

If there are no ties, an equal but simplified formula is

$$r_{s,STAT} = 1 - \frac{6\sum d_i^2}{n(n^2-1)},$$

where $d = x - y$.

Why?

Hence we can rewrite the denominator:

$$\frac{\sum_i (x_i-\bar{x})(y_i-\bar{y})}{\sum_i (x_i-\bar{x})^2}$$

But the denominator is just a function of n:

$$\begin{aligned}\sum_i (x_i-\bar{x})^2 &= \sum_i x_i^2 - n\bar{x}^2\\ &= \frac{n(n+1)(2n+1)}{6} - n(\frac{(n+1)}{2})^2\\ &= n(n+1)(\frac{(2n+1)}{6} - \frac{(n+1)}{4})\\ &= n(n+1)(\frac{(8n+4-6n-6)}{24})\\ &= n(n+1)(\frac{(n-1)}{12})\\ &= \frac{n(n^2-1)}{12}\end{aligned}$$

$$\begin{aligned}\sum_i (x_i-\bar{x})(y_i-\bar{y}) &= \sum_i x_i(y_i-\bar{y}) - \sum_i \bar{x}(y_i-\bar{y})\\ &= \sum_i x_i y_i - \bar{y}\sum_i x_i - \bar{x}\sum_i y_i + n\bar{x}\bar{y}\\ &= \sum_i x_i y_i - n\bar{x}\bar{y}\\ &= \sum_i x_i y_i - n(\frac{n+1}{2})^2\\ &= \sum_i x_i y_i - \frac{n(n+1)}{12}3(n+1)\\ &= \frac{n(n+1)}{12}.(-3(n+1)) + \sum_i x_i y_i\\ &= \frac{n(n+1)}{12}.[(n-1)-(4n+2)] + \sum_i x_i y_i\\ &= \frac{n(n+1)(n-1)}{12} - n(n+1)(2n+1)/6 + \sum_i x_i y_i\\ &= \frac{n(n+1)(n-1)}{12} - \sum_i x_i^2 + \sum_i x_i y_i\\ &= \frac{n(n+1)(n-1)}{12} - \sum_i (x_i^2+y_i^2)/2 + \sum_i x_i y_i\\ &= \frac{n(n+1)(n-1)}{12} - \sum_i (x_i^2 - 2x_i y_i + y_i^2)/2\\ &= \frac{n(n+1)(n-1)}{12} - \sum_i (x_i-y_i)^2/2\\ &= \frac{n(n^2-1)}{12} - \sum d_i^2/2\end{aligned}$$

Numerator/Denominator

$$\begin{aligned}&= \frac{n(n+1)(n-1)/12 - \sum d_i^2/2}{n(n^2-1)/12}.\\ &= \frac{n(n^2-1)/12 - \sum d_i^2/2}{n(n^2-1)/12}\\ &= 1 - \frac{6\sum d_i^2}{n(n^2-1)}\end{aligned}$$

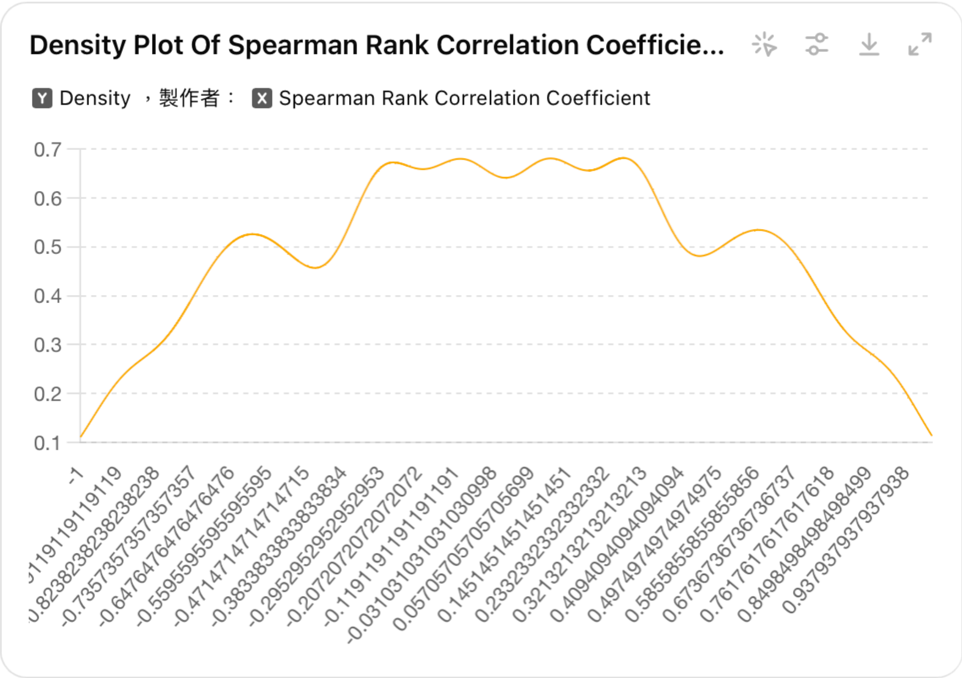
Hence

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}.$$

- [https://stats.stackexchange.com/questions/8912... equivalence-of-the-following-two-formulas-for-... correlation]

SAMPLING DISTRIBUTION OF $r_{s,STAT}$

can you show me the density plot of the Spearman rank correlation coefficient when n = 5? thanks



Here is the density plot of the Spearman rank correlation coefficient for $n = 5$. If you have any further questions or need additional analysis, feel free to ask!

Critical Values

Table 9

■ **Table 9 Critical Values of Spearman's Rank Correlation Coefficient for a One-Tailed Test**

n	α = .05	α = .025	α = .01	α = .005
5	.900	—	—	—
6	.829	.886	.943	—
7	.714	.786	.893	—
8	.643	.738	.833	.881
9	.600	.683	.783	.833
10	.564	.648	.745	.794

Rejection region for a two-tailed test

Figure 15.12
Rejection region for a two-tailed test of the null hypothesis of no association, using Spearman's rank correlation test

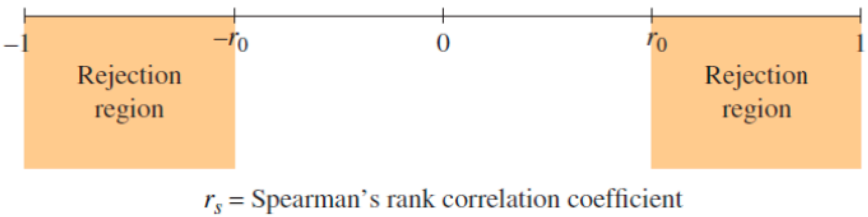


Table 9 Critical Values of Spearman's Rank Correlation Coefficient for a One-Tailed Test

n	$\alpha = .05$	$\alpha = .025$	$\alpha = .01$	$\alpha = .005$
5	.900	—	—	—
6	.829	.886	.943	—
7	.714	.786	.893	—
8	.643	.738	.833	.881
9	.600	.683	.783	.833
10	.564	.648	.745	.794
11	.523	.623	.736	.818
12	.497	.591	.703	.780
13	.475	.566	.673	.745
14	.457	.545	.646	.716
15	.441	.525	.623	.689
16	.425	.507	.601	.666
17	.412	.490	.582	.645
18	.399	.476	.564	.625
19	.388	.462	.549	.608
20	.377	.450	.534	.591
21	.368	.438	.521	.576
22	.359	.428	.508	.562
23	.351	.418	.496	.549
24	.343	.409	.485	.537
25	.336	.400	.475	.526
26	.329	.392	.465	.515
27	.323	.385	.456	.505
28	.317	.377	.448	.496
29	.311	.370	.440	.487
30	.305	.364	.432	.478

Source: From "Distribution of Sums of Squares of Rank Differences for Small Samples" by E.G. Olds, *Annals of Mathematical Statistics* 9 (1938).

Hypothesis test using Spearman's rank

1. H_0: No association between the rank pairs ($\rho_s = 0$) versus H_a: one or two-tailed alternative ($\rho_s \neq 0$, $\rho_s < 0$, $\rho_s > 0$)
2. Set α
3. Test statistic, Spearman's rank, and its distribution:

$$r_s \sim \text{Check Table 9}$$

4. Calculate realized r_s^*
5. Find rejection region or p-value using Table 9
6. Conclude.

RANK CORRELATION COEFFICIENT

H_0: no association between the rank pairs

H_a: one or two-tailed alternative

Test statistic: r_s ~ **Check Table 9**

Critical values for a one or two-tailed rejection region can be found using **Table 9** in Appendix I.

Rank the exam scores (the first variable is already in rank form). There are no ties.

IPLE

Teacher	1	2	3	4	5
Judge's Rank	4	2	3	1	5
Exam score	72	69	82	93	80

Teacher	1	2	3	4	5
x	4	2	3	1	5
y	2	1	4	5	3
d	**2**	**1**	**-1**	**-4**	**2**

For a one-tailed test with α = .05 and n = 5, reject if r_s <= -.900. We do not reject H_0. Not enough evidence to indicate a negative association.

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)} = 1 - \frac{6(26)}{5(25-1)} = \text{-}0.3$$

TABLE 9 Critical Values of Spearman's Rank Correlation Coefficient for a One-Tailed Test

n	$\boxed{\alpha = .05}$	$\alpha = .025$
5	.900	—
6	.829	.886
7	.714	.786
8	.643	.738
9	.600	.683

Raw data					
Teacher	1	2	3	4	5
Judges' rank	4	2	3	1	5
Exam score	72	69	82	93	80

Ranked data					
Teacher	1	2	3	4	5
Judges' rank	4	2	3	1	5
Exam score	2	1	4	5	3

Procedures in finding the best fitting regression line

- Calculate$(\sum x_i)$, $(\sum y_i)$, $(\sum x_i^2)$, $(\sum y_i^2)$,$(\sum x_i y_i)$, $\bar{x}$, and $\bar{y}$
- Calculate
  - $S_{xx} = \sum (x - \bar{x})^2 = \sum x^2 - (\sum x)^2/n$.
  - $S_{yy} = \sum (y - \bar{y})^2 = \sum y^2 - (\sum y)^2/n$.
  - $S_{xy} = \sum (x - \bar{x})(y - \bar{y}) = \sum xy - (\sum x)(\sum y)/n$.

n	5						
sum x	15						
sum y	15						
x^2	16	4	9	1	25	sum x^2	55
y^2	4	1	16	25	9	sum y^2	55
x*y	8	2	12	5	15	sum x*y	42
Sxy	-3						
Sxx	10						
Syy	10						
Spearmans' rank	-0.3						

METHOD 2

Raw data					
Teacher	1	2	3	4	5
Judges' rank	4	2	3	1	5
Exam score	72	69	82	93	80

Ranked data					
Teacher	1	2	3	4	5
Judges' rank	4	2	3	1	5
Exam score	2	1	4	5	3

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)} = 1 - \frac{6(26)}{5(25-1)} = -0.3$$

d	2	1	-1	-4	2
d^2	4	1	1	16	4
Spearman's rank	-0.3				

Rejection region

Because this is a left-tailed test (see H_a), the critical value is 0.900. Thus, the rejection region is $\{r : r < -0.900\}$.

■ Table 9 Critical Values of Spearman's Rank Correlation Coe cient for a One-Tailed Test

n	$\alpha = .05$	$\alpha = .025$	$\alpha = .01$	$\alpha = .005$
5	$\boxed{.900}$	—	—	—
6	.829	.886	.943	—
7	.714	.786	.893	—
8	.643	.738	.833	.881
9	.600	.683	.783	.833
10	.564	.648	.745	.794

HYPOTHESIS TEST OF SPEARMANS' RA… CORRELATION

1. H_0: No association between the judge's rank and the exam score ($\rho_s = 0$) versus H_a: There is agreement (negative association) between the judge's rank and the exam score ($\rho_s < 0$)
2. Set $\alpha = 0.5$.
3. Spearmans' rank: $r_{s,STAT} \sim$ check Tab 9
4. $r_s^* = -0.3$
5. Critical value 0.900. Rejection region is $\{r : r < -0.900\}$.
6. Do not reject H_0. No enough evidence to indicate a negative association.

It is likely that there is no association betweenjudge's rank and exam score!

Example: log Relationship